RAMYA AKULA, PH.D.

AI Research Scientist | Trustworthy AI · NLP · Computational Biology

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PROFESSIONAL SUMMARY

Ph.D.-trained AI researcher and educator with 10+ years spanning interpretable machine learning, natural language processing, computational biology, academic research, teaching, and international engineering. Developed peer-reviewed methods in interpretable AI and semantic evaluation, including work associated with U.S. Patent 12,210,833 and DARPA Forward Riser recognition. Recent postdoctoral research applies reproducible single-cell transcriptomic analysis to lung aging and disease. Experienced in translating complex scientific and engineering questions into rigorous computational workflows, interpretable results, and collaborative research outputs.

RESEARCH & TECHNICAL STRENGTHS

Interpretable & Trustworthy AI
Interpretability, evaluation, explainable ML, human-centered AI

NLP & Language Evaluation
Transformers, summarization, semantic similarity, figurative-language analysis

Computational Biology
scRNA-seq, Scanpy, Seurat, Harmony, differential expression, enrichment

Machine Learning
Deep learning, representation learning, transfer learning, statistical evaluation

Scientific Computing
Python, R, PyTorch, TensorFlow, Jupyter, Git, Linux, Docker

Research Translation
Scientific communication, mentoring, customer discovery, interdisciplinary collaboration

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar - Computational Biology & Machine Learning — **2024-2025**

The Ohio State University, Department of Internal Medicine (Pulmonary) | Columbus, OH

- Developed reproducible pipelines to integrate multi-cohort single-cell RNA-sequencing datasets, including quality control, Harmony-based batch correction, cell-type annotation, differential expression, and pathway-level interpretation.
- Analyzed transcriptomic signatures of lung aging, cellular senescence, apoptosis, donor type, and chronic lung injury across human and mouse cohorts.
- Partnered with pulmonary scientists to translate biological questions into analytical plans, interpretable visualizations, and publication-ready findings.
- Contributed to a 2026 peer-reviewed publication in The Journals of Gerontology.

Lecturer - Computer Science & M.S. FinTech — **2023-2024**

University of Central Florida | Orlando, FL

- Designed and taught graduate and undergraduate courses in artificial intelligence, NLP, machine learning, data science, algorithms, blockchain, smart contracts, and programming.
- Mentored approximately 80 students across applied AI, data-science, research, and software-development projects.
- Advised student teams on project scope, methodology, execution, and technical communication.

Graduate Research Assistant - Trustworthy AI & NLP — **2018-2022**

University of Central Florida, Complex Adaptive Systems Laboratory | Orlando, FL | Advisor: Dr. Ivan Garibay

- Developed an interpretable multi-head self-attention architecture for sarcasm detection; the work received a U.S. patent and DARPA Forward Riser recognition.
- Proposed SPEED Score, a sentence-embedding-based metric for evaluating abstractive and extractive summarization.
- Researched explainable AI, figurative-language understanding, toxic-language detection, semantic evaluation, and AI for social good under DARPA-supported programs.

- Served as NSF I-Corps Entrepreneurial Lead, conducting customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.

Project Engineer - Airbus A350 Program **2015-2017**

Umlaut GmbH (formerly P3 Group), Client: Airbus GmbH | Hamburg, Germany

- Supported the Airbus A350 product-structure transformation program through stakeholder meetings, requirements gathering, estimation, milestone planning, and engineering coordination.
- Developed tooling to reduce computational overhead in product-structure workflows and improve execution efficiency.

Graduate Research Assistant - Machine Learning & Visualization **2014-2015**

Technical University of Kaiserslautern | Germany

Software Quality Assurance Analyst - Client: Roche GmbH **2013-2014**

Exco GmbH | Germany

Graduate Research Assistant - AI Research **2013**

DFKI GmbH | Germany

Software Engineer **2011-2013**

Osmania University | India

Software Developer - Client: UnitedHealthCare **2010-2011**

Inventurus Software Solutions | India

SELECTED RESEARCH, INNOVATION & RECOGNITION

- U.S. Patent 12,210,833 (2025): Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection.
- DARPA Forward Riser (2022): Recognition associated with research in interpretable machine learning for figurative-language understanding.
- NSF I-Corps Entrepreneurial Lead (2019): Customer discovery and commercialization activities for an AI-assisted qualitative-analysis platform.
- First Place, Startup Weekend Orlando (2021): Contributed to an AI-powered online youth-safety product concept.
- Research has received coverage from outlets including NBC News, IEEE Spectrum, Los Angeles Times, Defense One, ScienceDaily, and ABC Australia, as listed in the prior CV.

SELECTED PUBLICATIONS

- Akula, R., et al. “Transcriptomic shifts in human lungs donated after brain and circulatory death and severe Obliterative Bronchiolitis.” The Journals of Gerontology, 2026.
- Akula, R., & Garibay, I. “SPEED Score: Sentence Pair Embeddings Based Evaluation Metric for Abstractive and Extractive Summarization.” LREC, 2022.
- Akula, R., & Garibay, I. “Interpretable Multi-Head Self-Attention Architecture for Sarcasm Detection in Social Media.” Entropy, 2021.
- Akula, R., & Garibay, I. “Explainable Detection of Sarcasm in Social Media.” ACL Workshop, 2021.

  Complete publication record on Google Scholar

TECHNICAL EXPERTISE

Natural Language Processing & Language Evaluation: Transformers, text summarization, semantic similarity, sentence embeddings, figurative-language understanding, toxicity detection, model evaluation.

Trustworthy AI & Machine Learning: Interpretability, explainable machine learning, deep learning, representation learning, transfer learning, statistical evaluation, human-AI collaboration.

Computational Biology: Single-cell RNA sequencing, Scanpy, Seurat, Harmony, differential expression, cell annotation, gene-set enrichment, transcriptomics.

Data & Scientific Computing: Python, R, PyTorch, TensorFlow, Jupyter, Git, Linux, Docker, high-performance computing, reproducible workflows.

EDUCATION

Ph.D. in Computer Science **2018-2022**
University of Central Florida | Orlando, FL | Advisor: Dr. Ivan Garibay

M.S. in Computer Science **2013-2015**
Technical University of Kaiserslautern | Germany | Advisor: Dr. Hans Hagen

B.Tech. in Computer Science and Engineering **2006-2010**
Jawaharlal Nehru Technological University | India